

South Carolina K–5 Vertical Articulation

Three-Dimensional Science Progression (SEPs, DCIs, CCCs) — 2021 CCR Standards

State: South Carolina Department of Education

Framework: 3D Science Learning (3-Dimensional)

Grade Scope: Elementary (K–5) Progression

South Carolina’s College- and Career-Ready Science Standards (2021) are structured around three distinct, integrated dimensions: **Science and Engineering Practices (SEPs)**, **Disciplinary Core Ideas (DCIs)**, and **Crosscutting Concepts (CCCs)**.

1. Dimension 1: Science & Engineering Practices (SEPs) Vertical Progression

SEP Practice Dimension	Grades K–2 Progression Baseline	Grades 3–5 Advanced Progression
1. Asking Questions & Defining Problems	Ask questions based on observations to find more information about the natural or designed world.	Ask questions that can be investigated empirically and predict outcomes based on patterns or relationships.
2. Developing & Using Models	Develop or use concrete models (drawings, physical replicas, dioramas) to represent events or design solutions.	Develop diagrams, physical models, or simulations to describe phenomena or test cause-and-effect relationships.
3. Planning & Carrying Out Investigations	Plan and conduct simple collaborative investigations to gather evidence or answer questions.	Plan and conduct investigations collaboratively, controlling variables and identifying data needed as evidence.
4. Analyzing & Interpreting Data	Record observations using pictures, numbers, or simple graphs to identify patterns in data.	Represent data in tables and graphical displays (bar graphs, line plots) to reveal patterns, trends, and rates of change.
5. Mathematics & Computational Thinking	Use counting and standard units to measure length and quantify physical properties.	Organize numerical data to compare properties, calculate averages, and construct simple algorithms.
6. Constructing Explanations & Solutions	Use evidence to construct simple accounts of natural phenomena or solve engineering design problems.	Construct explanations using evidence (measurements, observations) and design multiple solutions to problems.
7. Engaging in Argument from Evidence	Identify arguments that are supported by evidence versus those that are personal opinions.	Construct and critique arguments supported by empirical evidence, models, or scientific principles.
8. Obtaining, Evaluating, & Communicating	Read emergent texts and use oral/written language or drawings to communicate scientific findings.	Read and summarize grade-appropriate scientific texts; communicate data using tables, diagrams, and written reports.

2. Dimension 2: Disciplinary Core Ideas (DCIs) Vertical Scope (K–5)

Grade	Physical Science (PS)	Life Science (LS)	Earth & Space Science (ESS)
Kindergarten	Forces: Pushes and Pulls; Motion	Basic Needs of Plants & Animals	Weather Patterns & Human Environment Impact
1st Grade	Light & Sound Vibrations	Plant & Animal Structures, Traits, Growth	Sun, Moon, and Star Patterns in Space
2nd Grade	Properties & States of Matter	Ecosystems: Pollination & Seed Dispersal	Changes to Earth's Landforms; Water Bodies
3rd Grade	Balanced & Unbalanced Forces; Motion	Life Cycles, Traits Inheritance, Ecosystem Adaptation	Weather, Climate Regions, Natural Hazards

Grade	Physical Science (PS)	Life Science (LS)	Earth & Space Science (ESS)
4th Grade	Energy Transfer & Motion; Waves	Internal/External Plant & Animal Structures	Earth's Features, Mapping, Erosion, Renewable Energy
5th Grade	Structure & Properties of Matter (Particles)	Matter & Energy Flow in Ecosystems (Food Webs)	Earth Systems Interconnectedness; Stars & Solar System

3. Dimension 3: Crosscutting Concepts (CCCs) K–5 Growth

Crosscutting Concept	K–2 Conceptual Expectation	3–5 Conceptual Expectation
1. PATTERNS	Recognize and classify observed patterns in nature and human-designed world.	Use patterns to sort/classify, analyze rates of change, and predict natural phenomena.
2. CAUSE AND EFFECT	Events have causes that generate observable patterns; conduct simple testing.	Routinely identify, test, and use cause-and-effect relationships to explain changes.
3. SCALE, PROPORTION, & QUANTITY	Compare relative sizes/scales (bigger/smaller, faster/slower) using standard units.	Recognize natural objects exist from micro-scale to immensely large time and space spans.
4. SYSTEMS & SYSTEM MODELS	Understand that objects/organisms are made of smaller parts working together.	Describe a system in terms of its components and inputs/outputs to predict behaviors.
5. ENERGY & MATTER	Observe that objects can be changed by heating, cooling, or physical force.	Track energy transfer and conservation of matter during physical/biological changes.
6. STRUCTURE & FUNCTION	Observe that the shape/structure of an object or organism relates to its function.	Analyze how substructures of plants, animals, or engineered objects support specific roles.
7. STABILITY & CHANGE	Observe that some things stay the same while others change over time.	Measure rates of change and explain factors that cause stability or disruption in systems.

Implementation Note for SC Educators: Under the 2021 South Carolina Standards, a Performance Expectation (PE) is formed at the intersection of all three dimensions. Formative and summative assessments must evaluate students' ability to perform a Science Practice (SEP) using a Crosscutting Concept (CCC) to demonstrate understanding of a Disciplinary Core Idea (DCI).